

Karnataka Gravity Inversion: Comparing Regularizations

The same Bouguer gravity data on the same mesh, inverted at full resolution on AWS EC2 with six regularization settings. This report compares their data fit, lateral structure and depth, cross-checks them against the 24 September SimPEG run on an 8 km deep mesh and against Tomofast-x, and tests whether a coarse 2 km study can stand in for full-resolution runs.

AREA	DATA
70 × 70 km, easting 641–711 km, northing 1634–1704 km (UTM 43N, EPSG:32643)	NGPM Bouguer anomaly, 500 m grid decimated to 1 km: 5,040 points; second-order trend surface removed
MESH	RUNS
1 km × 1 km × 500 m core, 10 km deep; about 20 km padding at the sides and bottom (factor 1.3): 84 × 84 × 27 = 190,512 cells	6 full-resolution inversions on AWS EC2 (ap-south-1), 2 of them submitted from the upload page; a 16-run 2 km study run locally

Summary

- **All six runs fit the data about equally well.** χ^2/N 0.86–1.08, residual RMS 0.46–0.52 mGal. The fit alone cannot choose between these models.
- **The lateral structure is robust.** The vertically integrated density maps correlate at 0.83–0.98 pairwise, and at no less than 0.89 without the original settings. The NW–SE dense belt and the low-density belts to the east have the same position and strike in every model.
- **The regularization sets the depth.** Centroid depth under the main Bouguer high: 3.4 km for $\beta = 0.5$, 3.8 km for L1–L2, 5.6 km for $\beta = 1$, 6.4 km for $\alpha_s = 0.1$ and 7.6 km for $\beta = 1.5$. The original settings give a column from the surface to the bottom of the mesh (19.9 km). A larger β gives a deeper model with no matching change in χ^2/N .
- **A different set-up gives the same depth.** The 24 September SimPEG run on an 8 km deep mesh ($\alpha_s = 0.05$, sensitivity weighting) put the main body at 3.3–7.3 km (half maximum), centroid 5.3 km; this report's $\beta = 1$ run gives 3.3–7.8 km and 5.6 km. Tomofast-x fitted the data more closely (chi 0.65) but its model has no base, and it took about 2.7 hours against SimPEG's 9 minutes.
- **The coarse 2 km study is good for screening.** For three values of β it predicted the full-resolution centroid of the main body to within 0.6 km; for $\alpha_s = 0.1$ it was 1.5 km off. A coarse run takes about 25 s, a full-resolution run about 5 minutes.
- **Recommendation.** $\alpha_s = 1$ (now the pipeline default) with Li & Oldenburg depth weighting, β between 0.5 and 1, reporting a range of depths rather than a single model: $\beta = 0.5$ agrees with L1–L2, $\beta = 1$ with the 8 km deep mesh.

- **Against the published geology** (Section 6). The dense belt coincides with the mapped Sandur schist belt, and the four iron-ore mines of the belt lie on it. The depth range brackets the one published estimate, a basin about 6 km deep ([Maurya et al. 2023](#)).
- **Caveat.** Gravity data alone do not determine depth; the depths in this report are a consequence of the regularization. The actual depths need constraints from rock densities, geological sections or seismic data.

1 Data and the runs compared

Within the area the Bouguer anomaly ranges from -94.9 to -44.3 mGal. Its main feature is a NW–SE trending high from (650, 1685) to (675, 1660) km. After removing a least-squares second-order trend surface (the regional field, -95.3 to -75.9 mGal), the residual anomaly ranges from -16.4 to $+31.7$ mGal; this is what was inverted (Figure 1). Every point was given an error of 0.5 mGal. Bouguer data are positive downward and SimPEG's g_z is positive upward, so the pipeline multiplies the data by -1 before the inversion and returns the observed and predicted data in the original convention.

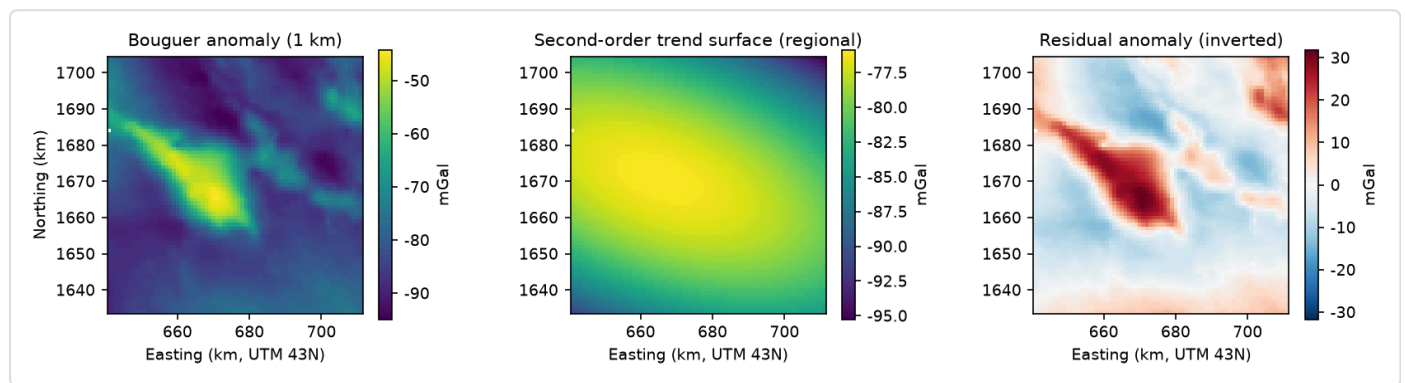


Figure 1 Left: Bouguer anomaly in the area (decimated to 1 km). Centre: least-squares second-order trend surface, removed as the regional field. Right: the residual anomaly, i.e. the data inverted (red is positive, dense).

The six runs share the mesh, the data and the errors; only the regularization differs:

Run	Regularization	α_s	Norms $p = [p_s, p_x, p_y, p_z]$	Depth weighting	Submitted from
Original sparse ($\alpha_s=1e-4$, sensitivity)	sparse (IRLS)	0.0001	[0, 2, 2, 1]	sensitivity	script ec2_multi_run.py
L1–L2 (IRLS)	elastic net (Utsugi 2019), L1 share 0.8, solved by IRLS	—	—	sensitivity	script ec2_multi_run.py
sparse $\alpha_s=1$, $\beta=0.5$	sparse (IRLS)	1	[0, 2, 2, 2]	Li & Oldenburg, $\beta = 0.5$	upload page (sweep)
sparse $\alpha_s=1$, $\beta=1$	sparse (IRLS)	1	[0, 2, 2, 2]	Li & Oldenburg, $\beta = 1$	script ec2_multi_run.py
sparse $\alpha_s=1$, $\beta=1.5$	sparse (IRLS)	1	[0, 2, 2, 2]	Li & Oldenburg, $\beta = 1.5$	upload page (sweep)
sparse $\alpha_s=0.1$, $\beta=1$	sparse (IRLS)	0.1	[0, 2, 2, 2]	Li & Oldenburg, $\beta = 1$	script ec2_multi_run.py

Common settings: density contrast bounded to -0.2 to $+0.5$ g/cc (rock samples 2.52–3.40 g/cc, median 2.66); β chosen automatically for $\chi^2 = N$; at most 30 iterations and 30 IRLS iterations. $\alpha_x = \alpha_y = \alpha_z = 1$, passed to SimPEG as length scales. Li & Oldenburg depth weighting weights the model norm by $(z + z_0)^{-\beta/2}$, where z is the depth below the nearest station and z_0 half the smallest cell.

2 Comparing the full-resolution results

2.1 Data fit

All six runs reached the target fit, with nearly the same statistics. The residual maps (Figure 2) differ, though:

- **Original settings:** the residuals cluster at a few places with steep gradients, as patches of positive and negative misfit up to 7.7 mGal. A model made of columns struggles to fit adjacent narrow anomalies.
- **L1–L2:** a broad positive residual over the main high, negative around it: the model underestimates the amplitude of the main anomaly (its main body is at the +0.5 g/cc bound). Its χ^2/N is also the highest (1.08).
- **The four depth-weighted runs:** the least structured residuals, mostly short-wavelength stripes along strike and a few isolated points. The isolated points near (660, 1680) and (702, 1659) km appear in every model and are probably outliers from the original gridding.

Run	χ^2/N	RMS (mGal)	Max residual (mGal)	Correlation, observed vs predicted
Original sparse ($\alpha_s=1e-4$, sensitivity)	0.97	0.49	7.7	0.9986
L1–L2 (IRLS)	1.08	0.52	4.2	0.9990
sparse $\alpha_s=1$, $\beta=0.5$	0.90	0.47	3.5	0.9989
sparse $\alpha_s=1$, $\beta=1$	0.96	0.49	4.0	0.9987
sparse $\alpha_s=1$, $\beta=1.5$	0.86	0.46	4.0	0.9989
sparse $\alpha_s=0.1$, $\beta=1$	0.89	0.47	5.2	0.9988

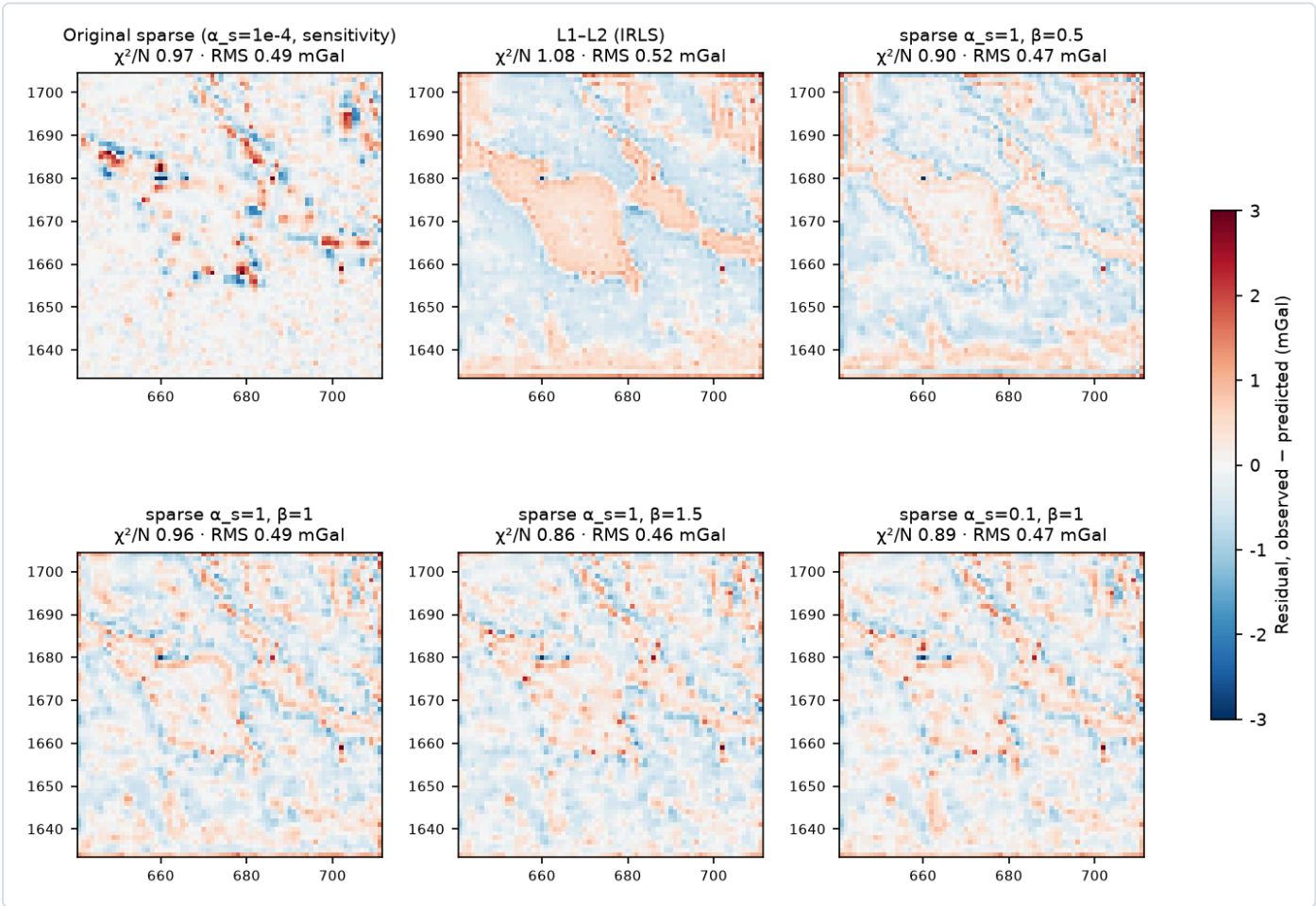


Figure 2 Residuals (observed – predicted) of the six full-resolution models; colour scale ± 3 mGal.

2.2 Lateral structure

Figure 3 shows five models at 1.8, 4.8 and 7.8 km depth. The top row of Figure 4 shows the vertically integrated density (the sum of density contrast $\times$ thickness in each column), the bottom row the centroid depth of each column.

- **The integrated density maps are nearly the same.** Pairwise correlations are 0.83–0.98, and no less than 0.89 without the original settings (table below). The NW–SE dense belt, the secondary high at its north-western end and the parallel low-density belts to the east have the same position and strike in every model.
- **What differs is the amplitude and sharpness.** A larger β gives stronger integrated density and sharper edges; the original settings give the smoothest model.
- **The centroid depth maps are completely different.** For the same dense belt the centroid lies at 2–4 km for $\beta = 0.5$ and L1–L2, 4–6 km for $\beta = 1$, 6–8 km for $\beta = 1.5$ (locally below 12 km), and deeper than 10 km for the original settings.
- **The slices show the same thing.** For $\beta = 0.5$ there is no anomaly left at 7.8 km; for $\beta = 1.5$ only scattered fragments remain at 1.8 km; the original model is almost the same at all three depths.

	Original	L1–L2	$\beta = 0.5$	$\beta = 1$	$\beta = 1.5$	$\alpha_s = 0.1$
Original	–	0.95	0.92	0.85	0.83	0.92
L1–L2	0.95	–	0.98	0.93	0.90	0.96
$\beta = 0.5$	0.92	0.98	–	0.93	0.89	0.95
$\beta = 1$	0.85	0.93	0.93	–	0.96	0.96
$\beta = 1.5$	0.83	0.90	0.89	0.96	–	0.94
$\alpha_s = 0.1$	0.92	0.96	0.95	0.96	0.94	–

Correlation between the vertically integrated density maps in the 70 $\times$ 70 km core. Green: ≥ 0.90 .

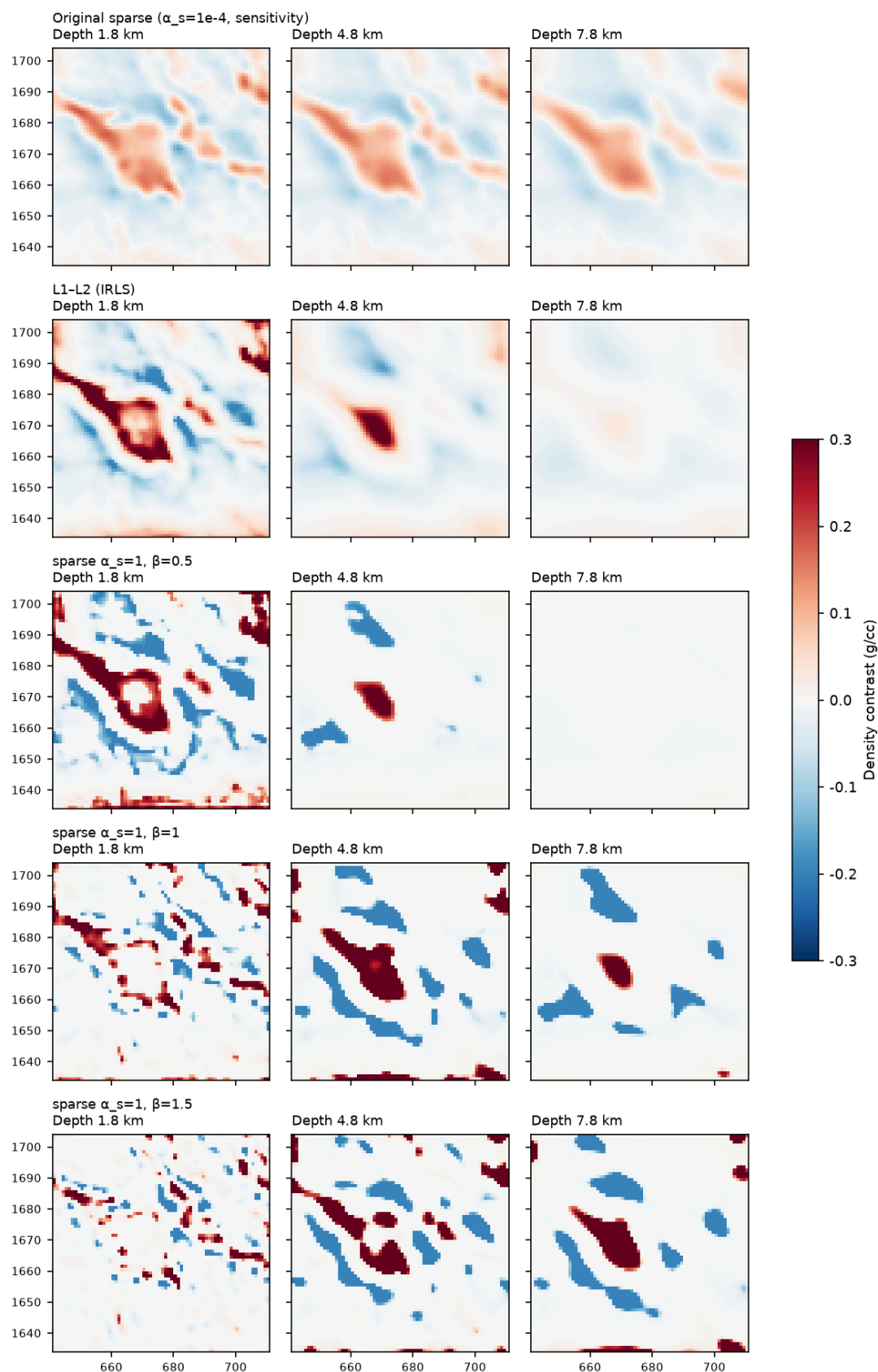


Figure 3 Depth slices of five models (core area only); colour scale ± 0.3 g/cc.

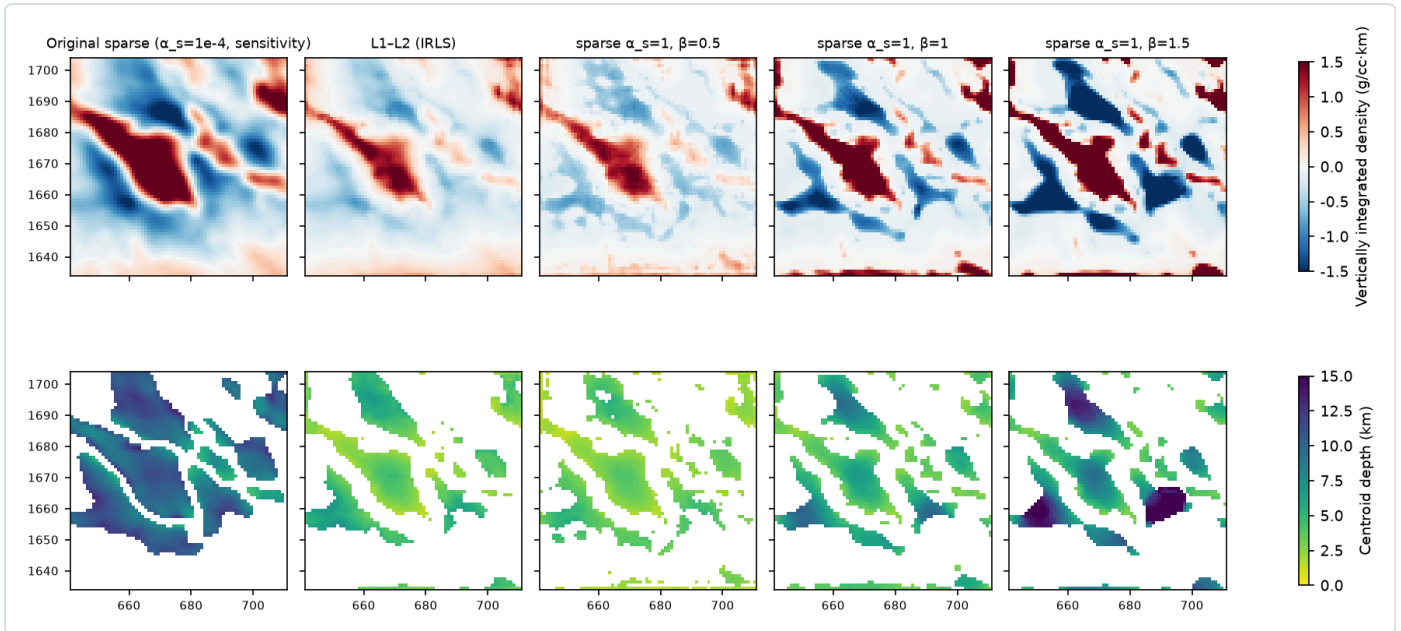


Figure 4 Top: vertically integrated density (g/cc-km); the five maps are essentially the same. Bottom: centroid depth of |density contrast| in each column (only columns whose integral exceeds 20% of the maximum); it changes strongly with the regularization.

2.3 Depth

Figure 5 shows an E–W section through the main Bouguer high (northing 1664.5 km), and the left panel of Figure 6 the distribution of |mass| with depth below the core area. The table gives where the mass lies, and measures of the density profiles under the main Bouguer high (easting 671.5 km, northing 1664.5 km) and the north-western high (650.5, 1682.5 km).

Depth is given as the **centroid** (the mean depth of the positive density contrast, weighted by thickness) rather than the depth of the peak: with $\alpha_s = 1$ the main body reaches the +0.5 g/cc bound, so the peak is a plateau with no single depth. The half-maximum range is the depth interval where the density exceeds half the peak.

Run	Core	Lateral padding	Below 10 km	D50 km	D90 km	Main high, half-max km	Main high, centroid km	NW high, half-max km	NW high, centroid km
Original sparse ($\alpha_s=1e-4$, sensitivity)	24%	26%	50%	9.1	17.6	0.3–19.9	10.1	0.3–10.3	8.1
L1–L2 (IRLS)	80%	9%	11%	3.0	9.2	2.3–4.8	3.8	1.3–2.8	2.2
sparse $\alpha_s=1$, $\beta=0.5$	86%	3%	11%	2.4	8.5	2.3–4.8	3.4	0.8–2.8	1.6
sparse $\alpha_s=1$, $\beta=1$	83%	3%	15%	4.6	10.5	3.3–7.8	5.6	1.3–4.3	2.8
sparse $\alpha_s=1$, $\beta=1.5$	60%	7%	33%	7.1	15.2	4.3–11.1	7.6	1.8–5.3	3.5
sparse $\alpha_s=0.1$, $\beta=1$	81%	6%	13%	5.4	9.8	3.8–9.3	6.4	0.3–4.8	2.7
SimPEG, 8 km deep mesh (24 Sep)	–	–	–	–	–	3.3–7.3	5.3	1.3–4.5	3.1

Core, lateral padding and below 10 km add up to 100%. D50, D90: half and 90% of the |mass| below the core area lie above this depth. Red: lateral padding > 20%, below 10 km > 30%, D50 > 7 km, or the main body reaching below 12 km. The 8 km deep mesh is a different mesh, so only the profile measures are comparable.

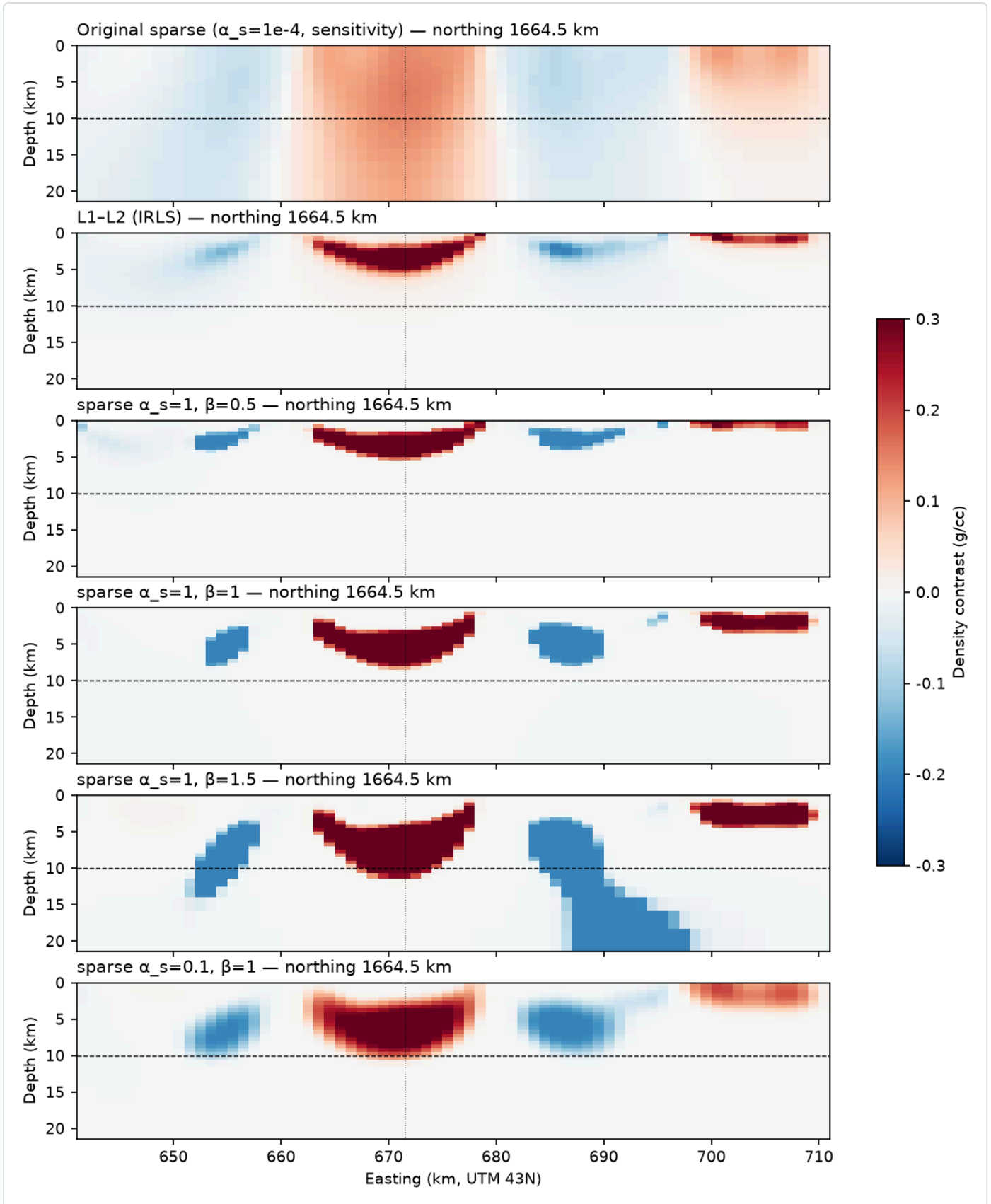


Figure 5 E–W sections through the main Bouguer high (northing 1664.5 km) in the six models. The vertical dotted line marks the high, the horizontal dashed line the base of the core mesh (10 km); below it is padding. Colour scale ± 0.3 g/cc.

- **The original settings give columns.** Under the main high the density contrast is only 0.15 g/cc and extends from 0.3 to 19.9 km, centroid 10.1 km; 50% of the mass lies below 10 km and 26% in the lateral padding.
- **The other five have a base.** The main body ends above 12 km, and at least 80% of the mass lies in the core (60% for $\beta = 1.5$).

- **β controls the depth monotonically.** For $\beta = 0.5, 1$ and 1.5 the centroid of the main body is at 3.4, 5.6 and 7.6 km, the half-maximum range 2.3–4.8, 3.3–7.8 and 4.3–11.1 km, while χ^2/N is 0.90, 0.96 and 0.86, unrelated to depth. This is the non-uniqueness of gravity inversion: the data cannot choose β .
- **L1–L2 is nearly the same as $\beta = 0.5$.** Two quite different regularizations give the same half-maximum range (2.3–4.8 km), with centroids 0.4 km apart.
- **$\alpha_s = 0.1$ lies between $\beta = 1$ and $\beta = 1.5$.** Centroid 6.4 km, peak 0.48 g/cc, close to but not at the bound. The model is smoother than with $\alpha_s = 1$; the north-western high extends from 0.3 to 4.8 km.
- **$\beta = 1.5$ starts to trail to depth.** The low-density body in the east reaches 20 km (Figure 5), and 33% of the mass lies below 10 km.
- **The north-western high is always shallower.** Apart from the original settings, its centroid lies at 1.6–3.5 km, 1.7–4.1 km above that of the main high. This relation holds in every model.

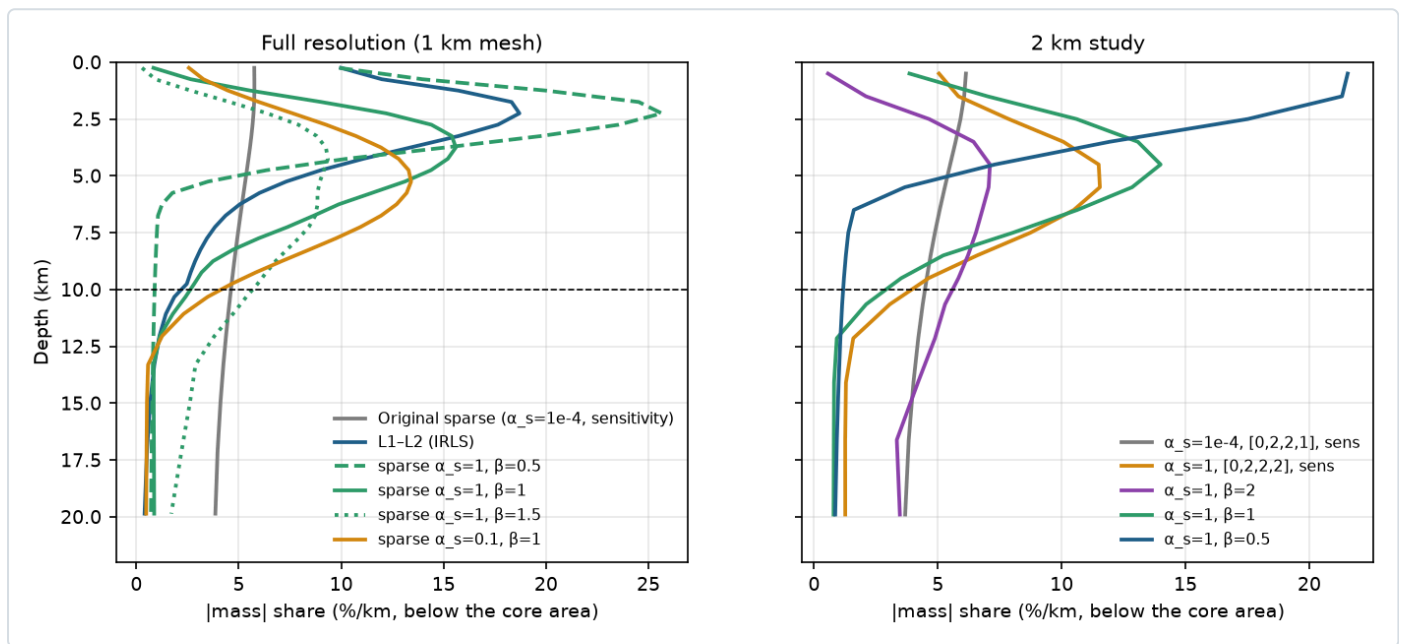


Figure 6 Distribution of $|mass|$ with depth below the core area (percent per km). Left: the six full-resolution models. Right: five runs of the 2 km study comparing α_s and depth weighting (Section 4). The dashed line is the base of the core mesh.

The radially averaged power spectrum of the residual anomaly (Figure 7) has three straight segments, giving mean depths to the top of the sources of 6.0, 2.2 and 1.1 km (Spector & Grant). It shows sources at several depths but cannot choose between the models: the top of the main body for $\beta = 0.5$ and L1–L2 (2.3 km) is near the intermediate estimate, the centroid for $\beta = 1$ (5.6 km) near the deep one. Spectral depths are averages over an ensemble of sources and are very uncertain.

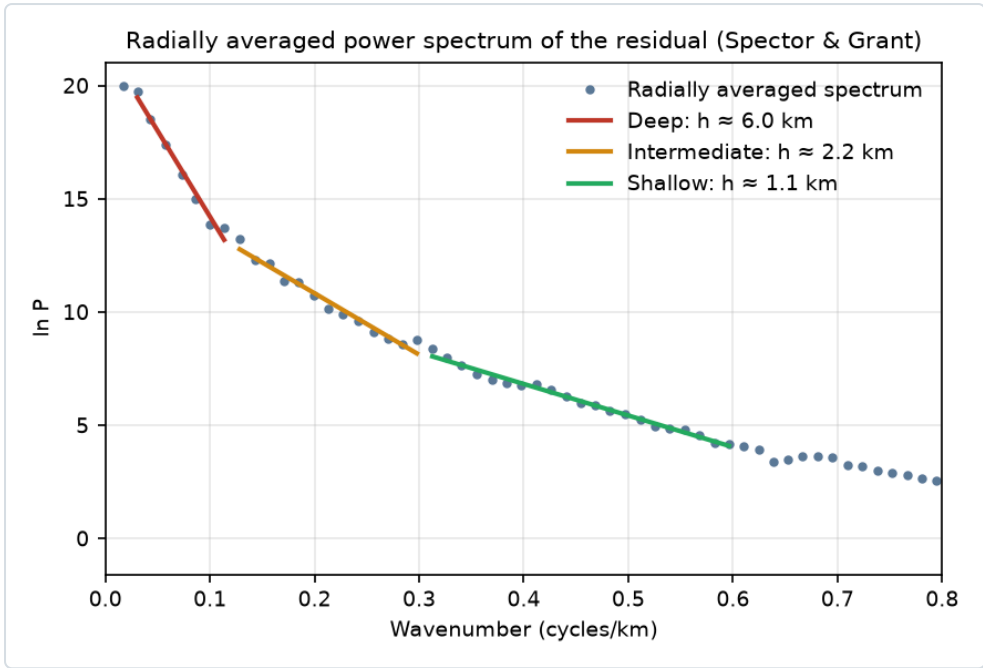


Figure 7 Radially averaged power spectrum of the residual anomaly (500 m grid, second-order trend removed). The slopes of the three segments give the mean depth to the top of the sources, $h = -\text{slope} / 4\pi$.

2.4 What is robust

The same under every setting:

- The lateral position, strike and extent of the anomalous bodies: the NW–SE dense belt and the low-density belts to the east.
- Sign and relative strength: the strongest positive body lies under the main Bouguer high.
- The main body has a base (in the five models other than the original settings, and in the independent 8 km deep mesh run); the north-western high is shallower than the main high.

Changes with the regularization:

- Absolute depth: the centroid of the main body lies at 3.4–7.6 km depending on β .
- Thickness and density contrast: with $\alpha_s = 1$ the main bodies reach the +0.5 / -0.2 g/cc bounds. This blocky, saturated model is a product of the regularization, not a measurement of density.
- Deep details, such as the low-density body in the east reaching 20 km for $\beta = 1.5$.

χ^2/N (0.86–1.08) cannot be used to choose between these models.

3 Cross-check: SimPEG on an 8 km deep mesh, and Tomofast-x

On 24 September the same Karnataka gravity data were inverted with two independent codes: SimPEG, used throughout this project, and Tomofast-x 2.0, an open-source parallel code. Both used the same 8 km deep mesh (500 m horizontal and 250 m vertical cells), the same cloud machine (16 vCPU, 123 GB memory) and the same noise model. Only the gravity part is summarised here; for magnetics see the original report (Desktop/Tomofastx_vs_SimPEG_package).

SimPEG	Tomofast-x 2.0	
Data fit, chi	1.04	0.65

	SimPEG	Tomofast-x 2.0
Data used	4,900 points (every second node)	19,600 points (all)
Under the main Bouguer high	3.3–7.3 km at half maximum, centroid 5.3 km, peak 0.64 g/cc; 0.03 g/cc at 9.2 km, while the mesh reaches 14.4 km	No base: the 1st, 50th and 99th percentiles of each layer change by less than 10% from the surface to 8 km
Regularization	Chosen once: sparse, norms (0, 2, 2, 1), $\alpha_s = 0.05$, sensitivity-based depth weighting	24-trial search of two smoothing weights; still no shape
Time to the final model	9 minutes (522 s)	About 2.7 hours (1.7 h search + 1.0 h full run)

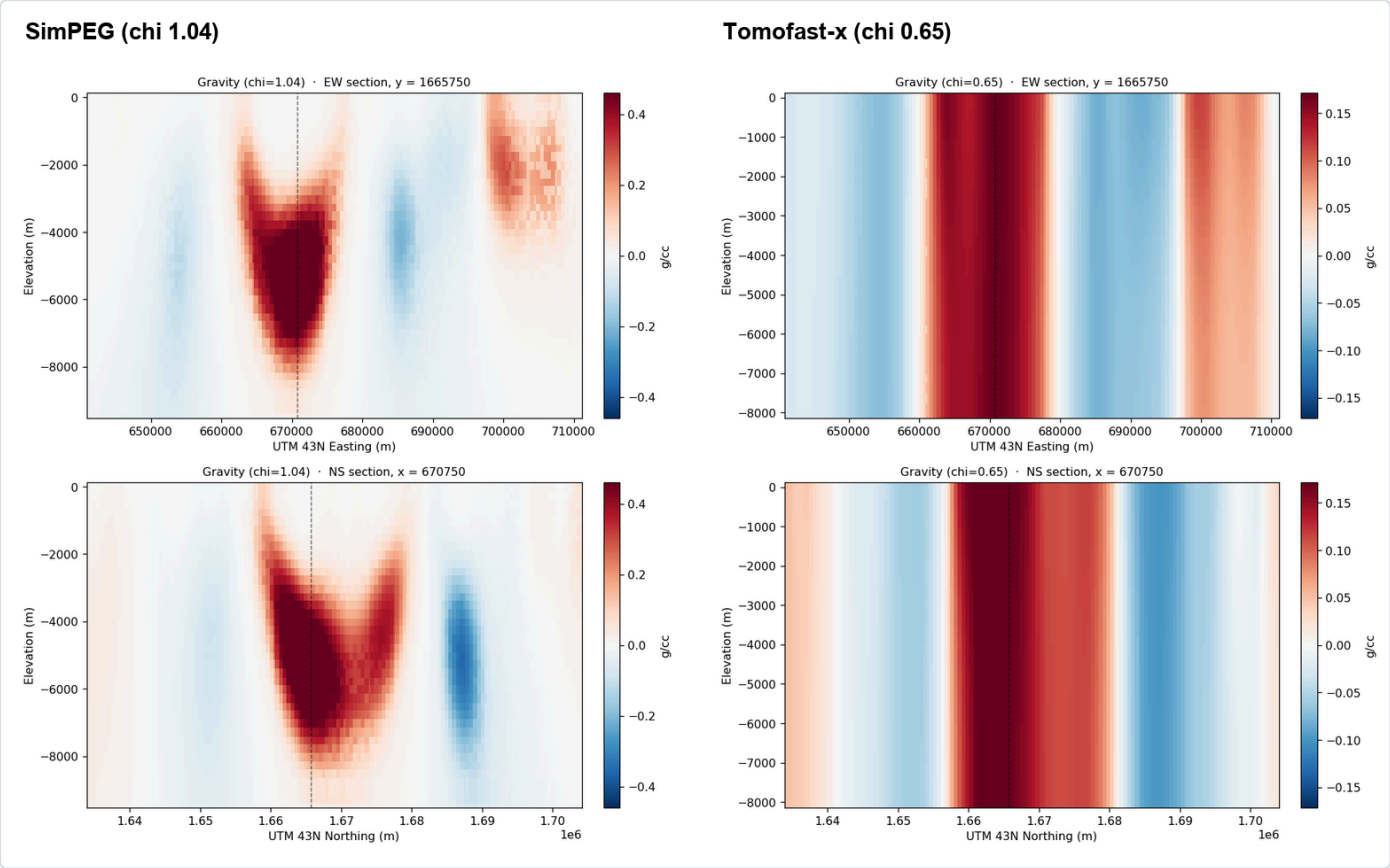


Figure 8 E–W and N–S sections through the anomaly centre (gravity only, from the original report). Left: SimPEG, a body closed at both ends. Right: Tomofast-x, vertical stripes from the surface to the mesh floor. Note the different colour scales.

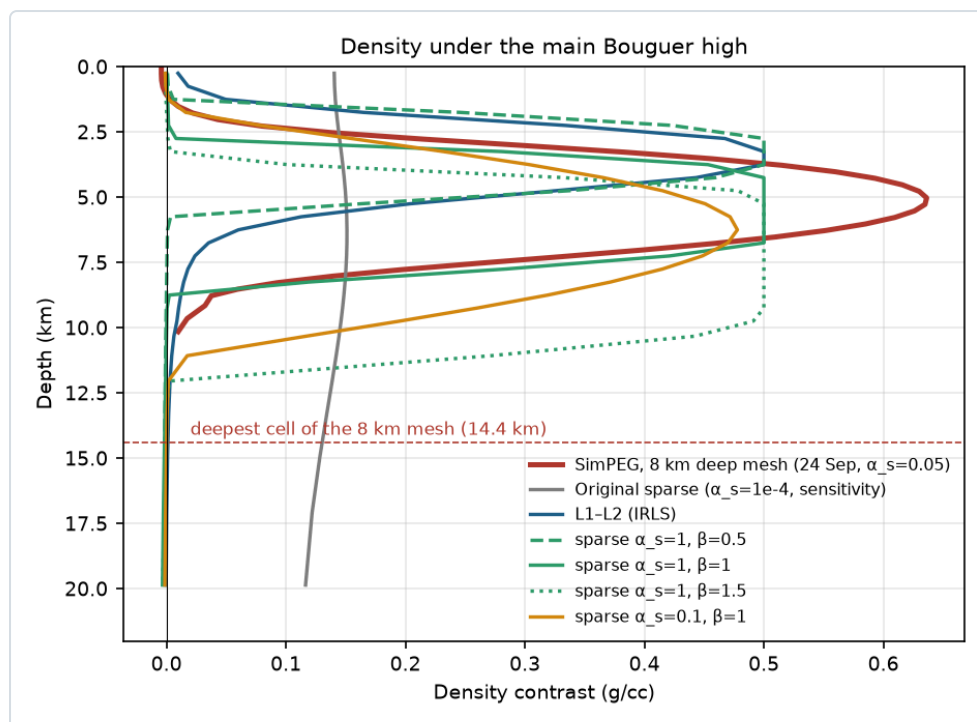


Figure 9 Density with depth under the main Bouguer high. Red: SimPEG on the 8 km deep mesh (24 September, no bounds); the others are this report's six full-resolution models. The dashed line is the deepest cell of the 8 km mesh.

How it relates to this report

- **Why the 24 September SimPEG model has a base.** It used $\alpha_s = 0.05$. With length-scale alphas the smallness term is then about a fifth of the gradient terms, so the compactness constraint acts; the original settings in this report used $\alpha_s = 10^{-4}$, about 2500 times weaker, and gave columns (Section 5).
- **Two different set-ups give the same depth.** The 8 km mesh (500 m cells, sensitivity weighting, 4,900 data) gives 3.3–7.3 km with the centroid at 5.3 km; the 1 km tensor mesh with $\alpha_s = 1$ and $\beta = 1$ (5,040 data) gives 3.3–7.8 km and 5.6 km (Figure 9). The north-western high agrees too (centroid 3.1 against 2.8 km). The result is therefore not sensitive to the mesh or the thinning, but both are the same kind of depth-weighted compact SimPEG model and do not remove the non-uniqueness: $\beta = 0.5$ and L1–L2 are shallower, $\beta = 1.5$ deeper, and the data cannot tell them apart.
- **A good chi is not evidence of a good model.** Tomofast-x reached chi 0.65, below the target, with no vertical shape at all; this report's β sweep also shows models with nearly the same chi whose depths differ by a factor of two. A settings search that looks only at chi selects physically meaningless settings. A published Tomofast-x application (Bird et al., 2025, Antarctic bathymetry) reached the same conclusion and ran 72 full-length trial inversions instead.

Cost (gravity only)

Stage	Approx. time
SimPEG gravity, final run (8 km mesh, 16 vCPU)	9 minutes (522 s)
Tomofast-x gravity: 24-trial settings search	1.7 hours
Tomofast-x gravity: full-length run	1.0 hour
SimPEG (this report, EC2 c5.2xlarge, 1 km mesh)	5–6 minutes per run, about \$0.03

Most of Tomofast-x's cost comes from one feature: it does not tune its own regularization, so its two smoothing weights must be found before a run by repeated trials, each 12–17 minutes on the full problem. SimPEG adjusts β during the run and needs no search.

What this does not show

- **SimPEG used less data:** 4,900 points against Tomofast-x's 19,600. Tomofast-x handles the full data volume with a wavelet-compressed sensitivity matrix, a real strength this comparison does not credit.
- **SimPEG had its own problems:** two faults in the workflow at the time (a mesh-depth setting that had no effect, a missing solver speed-up) took hours to find. They are fixed, but the time is not in the table.
- **Tomofast-x can converge here:** on a shallower 3 km mesh it reached an on-target fit (gravity chi 1.87) with a search. The failure is in producing a bounded body at depth, not in fitting the data.
- **We may not have used Tomofast-x well:** its published applications also use bound constraints and other controls that we did not try. Its smoothness term cannot be made "blocky" as SimPEG's can, our best explanation for the copied-downward pattern, but that is a hypothesis and was not tested further.
- **One dataset, one operator:** a single survey handled by one person new to Tomofast-x.

4 The 2 km study against the full-resolution runs

Before the full-resolution runs, 16 sparse inversions were run locally on a 2 km × 2 km × 1 km mesh (every fourth point, 1,295 data), about 25 s each. The table lists all 16; the right panel of Figure 6 shows the depth distribution of five of them.

Run	α_s	Norms p	Weighting	χ^2/N	Core	Lateral padding	Below 10 km	D50 km	Main high, half-max km	Main high, centroid km
base	0.0001	[0,2,2,1]	sensitivity	1.00	25%	25%	50%	8.7	0.5–19.9	10.2
as1e-2	0.01	[0,2,2,1]	sensitivity	1.00	33%	21%	45%	8.2	0.5–19.9	9.7
as0.1	0.1	[0,2,2,1]	sensitivity	1.02	36%	30%	34%	6.7	0.5–9.5	5.5
as1	1	[0,2,2,1]	sensitivity	1.04	44%	33%	22%	5.9	3.5–8.5	6.3
as1_p0111	1	[0,1,1,1]	sensitivity	1.09	42%	35%	22%	6.0	3.5–8.5	6.2
as1_p0222	1	[0,2,2,2]	sensitivity	0.98	47%	33%	20%	5.3	3.5–8.5	6.2
as1_p0220	1	[0,2,2,0]	sensitivity	0.96	44%	32%	24%	5.8	3.5–7.5	5.8
as1e-4_p0222	0.0001	[0,2,2,2]	sensitivity	1.08	25%	26%	49%	8.3	0.5–19.9	9.4
as1_p0222_dw0.5	1	[0,2,2,2]	$\beta = 0.5$	0.97	80%	5%	15%	1.9	0.5–4.5	2.8
as1_p0222_dw1	1	[0,2,2,2]	$\beta = 1$	1.04	80%	5%	15%	4.6	3.5–7.5	5.8
as1_p0222_dw1.5	1	[0,2,2,2]	$\beta = 1.5$	1.00	58%	10%	32%	6.8	4.5–10.6	7.5
as1_p0222_dw2	1	[0,2,2,2]	$\beta = 2$	0.94	40%	6%	54%	8.9	5.5–14.1	9.7
as1_p0221_dw1	1	[0,2,2,1]	$\beta = 1$	0.94	77%	5%	18%	5.4	3.5–8.5	6.1
as1_p0111_dw1	1	[0,1,1,1]	$\beta = 1$	0.99	73%	9%	18%	5.4	3.5–8.5	6.0
as0.1_p0222_dw1	0.1	[0,2,2,2]	$\beta = 1$	1.03	68%	10%	21%	5.3	0.5–8.5	4.9
base_dw1	0.0001	[0,2,2,1]	$\beta = 1$	1.03	32%	19%	49%	8.6	0.5–19.9	10.2

"sensitivity" is SimPEG's sensitivity weighting, " $\beta = \dots$ " Li & Oldenburg depth weighting. Red cells as in Section 2.3.

Trends on the coarse mesh

- **α_s decides whether the model reaches the bottom of the mesh.** With sensitivity weighting, $\alpha_s = 10^{-4}, 10^{-2}, 0.1$ and 1 leave 50%, 45%, 34% and 22% of the mass below 10 km.
- **Depth weighting decides the lateral leakage.** With $\alpha_s = 1$, sensitivity weighting puts 33% of the mass in the lateral padding, depth weighting only 5%–10%.
- **β sets the mean depth.** $\beta = 0.5, 1, 1.5$ and 2 give $D50 = 1.9, 4.6, 6.8$ and 8.9 km and main-body centroids of 2.8, 5.8, 7.5 and 9.7 km. The $\beta = 2$ often used for gravity is the deepest here, with 54% of the mass below 10 km.
- **The norms matter less.** With $\alpha_s = 1$ and $\beta = 1$, $p = [0, 2, 2, 2], [0, 2, 2, 1]$ and $[0, 1, 1, 1]$ give main-body centroids of 5.8, 6.1 and 6.0 km.

Against the 1 km runs

At the same place (Figure 10), for the three values of β the coarse centroid is within 0.6 km of the 1 km result and the base within 0.5 km; the top is shallower on the coarse mesh (0.5 against 2.3 km for $\beta = 0.5$) because its cells are 1 km thick. $\alpha_s = 0.1$ is the exception: on the coarse mesh the body does not saturate (peak 0.20 g/cc), spans 0.5–8.5 km, and its centroid is 1.5 km shallower than at 1 km. The coarse mesh is good for screening β and comparing trends; the final result should still be confirmed at full resolution.

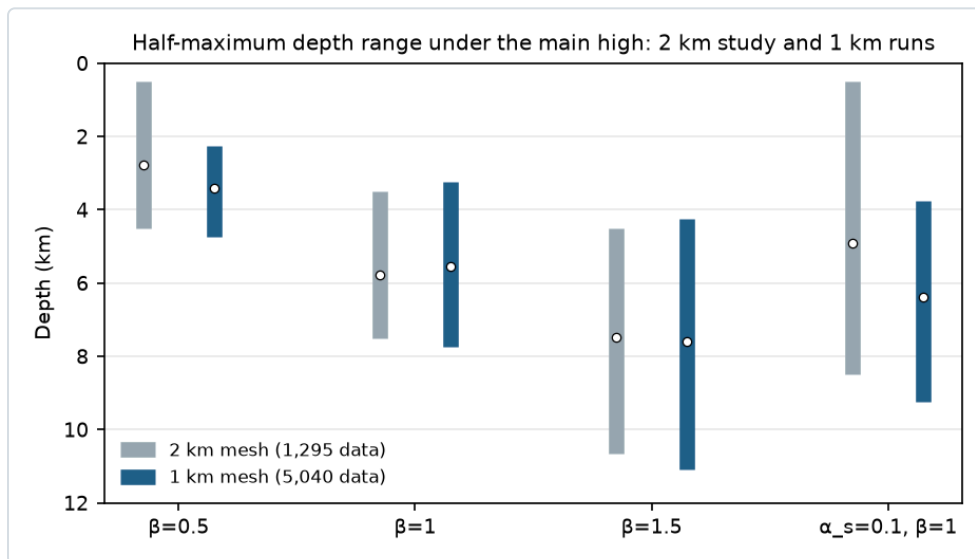


Figure 10 Half-maximum depth range (bars) and centroid (dots) under the main Bouguer high: the 2 km study (grey) and the 1 km full-resolution runs (blue).

5 Why the original settings give columns

The original settings were sparse (IRLS), $p = [0, 2, 2, 1]$, $\alpha_s = 10^{-4}$, $\alpha_x = \alpha_y = \alpha_z = 1$ (length scales), with sensitivity weighting. There are three reasons:

1. **The smallness term does almost nothing.** The pipeline passes $\alpha_{x,y,z}$ to SimPEG as length scales, so the gradient terms are weighted by $(1 \times 500 \text{ m})^2$, about $0.25 \Delta m^2$ per cell, some 2500 times stronger than the 10^{-4} m^2 of the smallness term; the compactness constraint $p_s = 0$ therefore has almost no effect.
2. **L1 on the vertical gradient.** $p_z = 1$ favours models that are piecewise constant with depth, so a column that does not change with depth costs almost nothing.
3. **Sensitivity weighting makes the padding cheap.** Cells outside the area and at depth have low sensitivity and therefore low weights, so mass placed there costs little (26% went into the lateral padding).

The 2 km study separates the three: changing only the depth weighting (`base_dw1` , α_s still 10^{-4}) leaves 49% of the mass below 10 km; changing only p_z to 2 (`as1e-4_p0222`) leaves 49%; changing only α_s to 1 brings it down to 22%, and $\alpha_s = 1$ with depth weighting to 15%. The first reason dominates. The pipeline default is now $\alpha_s = 1$.

6 Comparison with the published geology

ADDED 1 OCTOBER 2026 · LITERATURE CROSS-CHECK

The inversions of this report were run without reference to the geological map. This section compares their robust result, the position of the dense belt, with what is published about the area, and sets the depths of Section 2.3 against the one published estimate.

- **The dense belt is the Sandur schist belt.** The belt is mapped as a NW–SE greenstone belt about 60 km long and up to 18 km wide in its centre, enclosed by granite ([Mukhopadhyay & Matin 1993](#); [GSI, Karnataka](#)). The residual high of Figure 1 has the same length, width and strike, and its north-western end (the north-western high, 650.5, 1682.5 km) points at Hosapete (Hospet), where the belt is mapped to continue ([IJERT, Sandur Mn](#)).
- **The iron-ore mines sit on it.** Kumaraswamy, Donimalai, Ramandurg and NEB Range, which work the iron formation of the belt ([GSI, Karnataka](#); [MEAI 2024](#)), all lie on or at the edge of the residual high (Figure 11). The main Bouguer high (671.5, 1664.5 km) lies 4 km west of Donimalai and about 4.5 km north of the Kumaraswamy leases, in the south-eastern part of the belt where the published resource is concentrated: about 1,876 Mt of iron ore at about 63% Fe in the belt as a whole ([GSI, Karnataka](#)).
- **The densities agree.** Maurya et al. measured 2.86 to 3.56 g/cc on the mafic and ultramafic rocks of the belt and found the strong gravity anomalies over the schist belts, the weak ones over the granites ([Maurya et al. 2023](#)). This matches the samples used as bounds in the joint report (metabasalt 2.94 to 2.98 g/cc, iron formation 3.39 g/cc, granite about 2.63 g/cc).
- **The light belts are granite.** The low-density belts east of the dense belt lie where younger granites (2.5 to 2.6 Ga) override the eastern margin of the belt ([Mukhopadhyay & Matin 1993](#)).
- **The depth is consistent but not confirmed.** From a joint gravity–magnetic interpretation Maurya et al. describe the belt as a basin about 6 km deep ([Maurya et al. 2023](#)). That lies inside the range of this report: the main body ends at 4.8 km for $\beta = 0.5$, 7.8 km for $\beta = 1$ and 11.1 km for $\beta = 1.5$ (half maximum). Their depth also rests on an assumed density contrast, and the abstract gives no profiles, so the agreement shows that the β range of Section 7 is reasonable, not which β is right. The constrained joint inversion narrows it (base at about 5.5 km, joint report Section 5).
- **The shape inside the belt is not resolved by gravity.** The belt is traditionally read as a synclinorium; a structural study finds instead two metasedimentary belts on either side of a central metavolcanic terrane that do not join at either end, with steep bedding and folds plunging about 45° ([Mukhopadhyay & Matin 1993](#)). A single dense body, as in every model here, is compatible with either reading. The stratigraphy gives a weak preference, however. The iron formation is a layer at the top of the succession: the western sequence runs from metabasalt at the base to iron formation at the top ([Mukhopadhyay & Matin 1993](#)), a Bababudan-type assemblage of iron formation over mafic volcanics and quartzite ([GSI, Karnataka](#)). Folded steeply (near-isoclinal D1 folds), it stands as thin, near-vertical sheets on the margins of the belt, too thin to dominate the gravity, so the dense body of the models is more likely the metavolcanic core in the centre. That fits the two-flank reading better than a simple synform, whose youngest unit, the iron formation, would lie in its core.

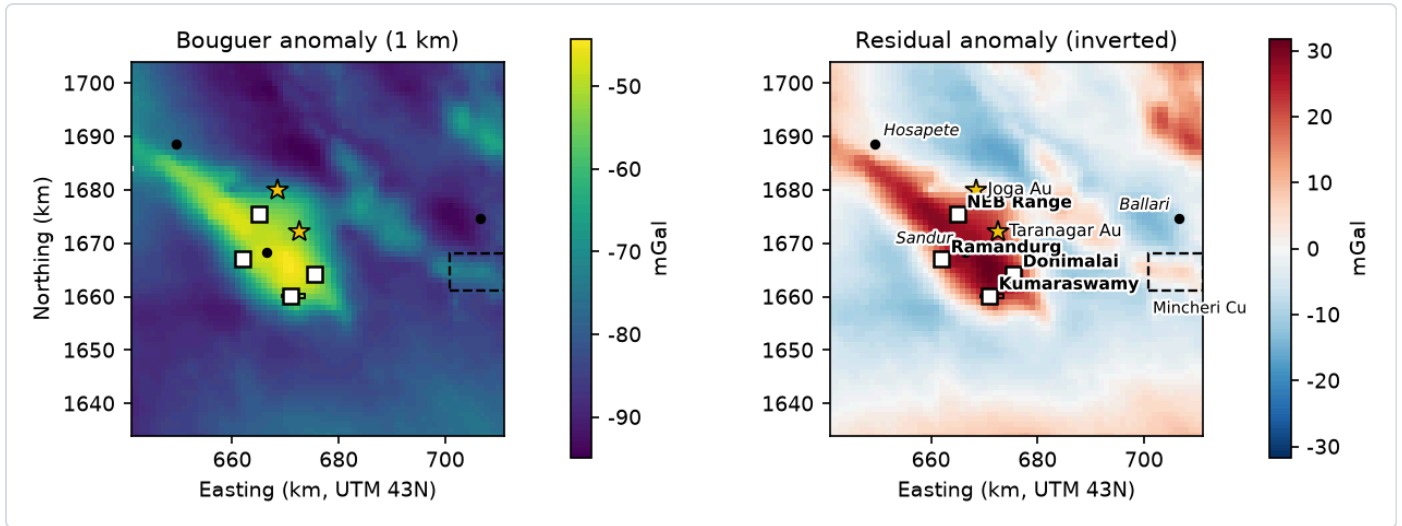


Figure 11 Published localities on the Bouguer anomaly (left) and the residual anomaly that was inverted (right). White squares: iron-ore mines (the two small rectangles beside Kumaraswamy are its B and C development blocks); yellow stars: gold occurrences; dashed box: the Mincheri copper reconnaissance block (clipped at the edge of the area); black dots: towns. Sources: [IBM 2024](#), [Kumaraswamy; IBM 2018](#), [NEB Range](#); [GEM.wiki](#); [Suresh & Basavanna 2014](#); [Sindhuja et al. 2020](#); [NMET](#), [Mincheri G4](#)).

Figure 12 sets the models against a schematic of the belt drawn from the DEM. Measured the same way, along the principal axes, the outline of its main belt is 42 km long and 16 km wide with a strike of 136° , and the residual anomaly above half its maximum (15.9 mGal) 43 km, 15 km and 131° (above 5 mGal it is 62 km long, about the mapped length). Leaving out a 5 km band along the edges of the area, 88% of the columns where the integrated density of the reference model (sparse $\alpha_s=1$, $\beta=1$) exceeds half its maximum lie inside the outline (84%–93% in the six models), which covers 12% of that area. Inside the outline the reference model averages 1.5 g/cc·km off the ridges and 0.7 g/cc·km on them; with 1 km cells and the ridges along the margins, any body centred in the belt would show this, so it is consistent with the reading above, not a test of it.

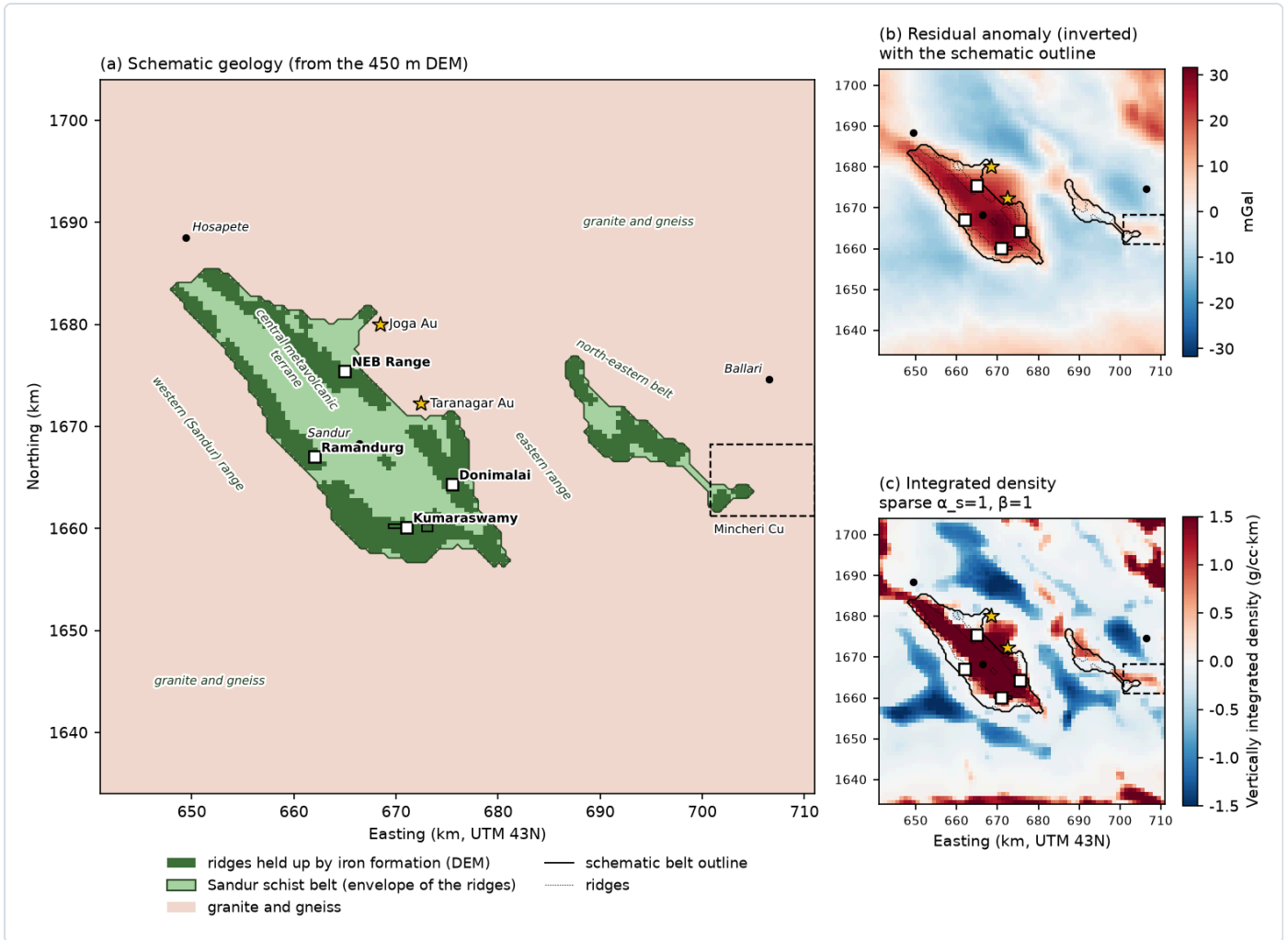


Figure 12 (a) A schematic of the belt drawn for this report from the 450 m DEM of the inputs, with the localities of Figure 11. Ridges (dark green) are where the ground stands more than 80 m above the median of a 12 km window (and above 600 m); they are read as the iron-formation ridges because the iron formation holds up the hills of the belt (MEAI 2024). The belt outline (light green) is the envelope of the ridges, and the central metamorphic terrane between the western and eastern ranges is placed after Mukhopadhyay & Matin (1993). It is not a geological map: the mapped belt is longer (about 60 km, continuing NW towards Hosapete) than the envelope (42 × 16 km). (b) The residual anomaly that was inverted (as in Figure 1), with the outline (solid) and the ridges (dotted) of panel a. (c) The vertically integrated density contrast of the reference model, sparse $\alpha_s=1$, $\beta=1$ (as in the top row of Figure 4), with the same outline. Symbols as in Figure 11.

For this report. The robust part of the gravity result, the position and strike of the dense belt, matches the mapped greenstone belt and its iron-ore district. The depth range of Section 2.3 brackets the one published estimate (about 6 km) but is not confirmed by it.

7 Recommendations

- Default settings:** sparse, $\alpha_s = 1$, $p = [0,2,2,2]$ or $[0,2,2,1]$, depth weighting `depth` with β between 0.5 and 1. Report the $\beta = 0.5$ and $\beta = 1$ models together as a depth range (main-body centroid about 3.4–5.6 km), with L1–L2 as an independent check.
- Coarse first, then fine:** screen settings on the 2 km mesh, confirm on the 1 km mesh. The upload page's parameter sweep submits several β at once and compares them in one workflow.
- Separate the robust from the rest when interpreting:** lateral position, strike and relative strength can be interpreted directly; depth, thickness and density contrast need independent constraints.

4. **Constrain depth with independent data:** rock densities, geological sections, boreholes or seismic data; or a gravity synthetic model resembling this area, to check which β recovers a known depth.
5. **Data:** remove isolated outliers or use a robust misfit, and check whether the 0.5 mGal noise floor is appropriate.
6. **Tomofast-x:** worth considering only where SimPEG exceeds memory even after thinning, or for joint gravity–magnetic inversion. If it is used, run full-length trials (not short proxies) and check the recovered depth profile, not chi alone.

Files

- This report and all figures: `examples/output/karnataka_gravity/scripts/` (`make_figures.py`, `build_reports.py`)
- The six full-resolution results: `data/ec2_runs/`; the 2 km study: `data/lowres_runs/`
- All runs in one interactive workflow (DAG viewer): `depth_study/karnataka_depth_study.geoinv3d_viewer.html`

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Mine and occurrence positions were converted to UTM 43N (EPSG:32643). Those taken from GEM.wiki and from latitude–longitude ranges in the papers are approximate, to about 1 km, and so is the outline of the Mincheri block. The Kumaraswamy and NEB Range coordinates are as printed in the IBM inspection reports, read as UTM 43N.

